

Continuous Distillation

Final Lab Report
Unit Operations Lab II
March 2nd, 2020

Group 2
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1. Executive Summary

The purpose of this experiment was to determine the efficiency of the binary distillation column separating ethanol and water while operating in a continuous steady state manner. Three experiments were performed: Calibration of the peristaltic feed pump, continuous distillation with a midpoint feed section, and then lastly continuous distillation with a feed position above the top plate of the column. For experiments B and C, two ethanol-water mixtures were used. The solutions consist of 10-liter volume of 10wt% ethanol and 90wt% water which was in the boiler, and the feed solution was 3-liter in which 30wt% was ethanol and 70wt% was water.

It was crucial to determine the accurate flow rate into the column when running at continuous distillation. For this reason, a calibration curved was performed on the peristaltic pump in order to pump 35mL of the 30wt% ethanol into the column. Based on the calibration curved data, 1.46 is the position of the variable speed dial of the pump in order to have a flowrate of 35mL/min. A refractometer calibration curved was also performed to show the relationship between refractive index and the weight percent of ethanol. This graph's trendline was used in calculating the composition of ethanol for samples in experiments B and C. The average composition of ethanol during the midpoint feed experiment was 0.637 in the distillate, and 0.136 in the bottoms. For the above top plate feed experiment the average composition of ethanol in the distillate was 0.6588 and 0.1392 in the bottoms. The average relative volatility was also calculated from the mole fractions for each experiment. The average relative volatility for the midpoint feed experiment was 1.853, while the volatility for the feed above the top plate was 2.028. The column has 8 actual trays, but when calculating the theoretical plates for each experiment it was found that the midpoint feed position had 2.9 plates, while the above feed plate had 2.506 theoretical plates. Lastly, the overall column efficiency for the midpoint feed was 36.36%, and the above top plate feed was 31.33%.

Beginning with the comparison of compositions of ethanol for each experiment, it was found that there was a larger composition of ethanol in the distillate for the above top plate feed position. Likewise, under the same experiment, it was found that the composition of ethanol in the bottoms was lower versus the midpoint composition. These results make sense, as feeding a more concentrated ethanol solution from the top, one would expect the concentration of ethanol will be greater at the top versus having a midpoint feed section. It was also found that the overall efficiency of the distillation column was greatest at the midpoint feed section. This was because the ideal location for the feed is somewhere where the feed stream composition is like the theoretical composition of the column. So, the middle feed location was better than the top location which resulted in the higher efficiency being in the middle.

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How was the McCabe-Thiele analysis?

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2. Introduction

The objective of this experiment is varying the feed position in a continuous operation distillation column and calculating the composition and overall column efficiency using Murphy's plate efficiency equation and Fenske's Equation. Also, the peristaltic feed pump had to be calibrated in order to deliver the correct amount of feed into the column, in which another objective was creating a calibration curve for this pump. The goal is to get the actual flow rate against the position of the variable speed dial. The significance of this experiment is to understand that distillation is classified as an equilibrium separation process since the gas and liquid phases are expected to reach phase equilibrium on each contacting stage of the distillation column. Also, it is very significant to see the relationship between the feed position of the column, and see which position has the greater overall column efficiency. This apparatus includes the feed tanks where the solution will be pumped from, then goes through a heat exchanger prior entering the column. The column also includes a condenser on top of the column, and a reboiler at the bottom. Common industrial applications can be observed when a chemical engineer faces a separation problem in process design, the usual first alternative is distillation. Industries such as oil refineries rely heavily on distillation columns in order to separate specific components to create the desired product for the customer.

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Also, mention some of the examples of the column being used in industry and use REFERENCES

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3. Theory

For distillation, the component with the lower boiling point will exit out of the top of the column as a gas. The component with the higher boiling point will exit out the bottom as a liquid. As the liquid begins to settle on a tray, heated gas will flow through it, which will increase the temperature and cause some of that liquid to turn into gas. On the reverse end, some of the liquid will fall back to the bottom, and liquify some of the gas. This process is called Vapor-Liquid Equilibrium (VLE).

The McCabe-Thiele model, figure 1 below, is used to determine the number of stages needed in a distillation column to achieve a desired separation. McCabe-Thiele plots are created by plotting the mole fraction of the most volatile component in the bottom vs the most volatile component in the distillate. Equilibrium data will be graphed, and so will the equation of equilibrium line (xy line) which is Henry's law: $Hx = y$. From the equilibrium line a line will be drawn horizontally until it reaches the xy line, or operating line, equation 1. Then, the line will be stepped back to the equilibrium line. This will be done until the entirety of the graph is filled, or when the desired equilibrium is obtained. Every step is considered one theoretical plate, which added up will yield the required plates to obtain the distillation. The reboiler and the condenser also count as stages. The issue with this model is that many assumptions must be made. For example, every tray is assumed to be in perfect equilibrium, the heats of vaporization for both the components in the feed are equivalent, and heat does not affect the solution. [2]

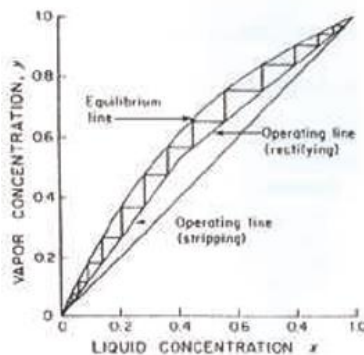


Figure 3.1: McCabe-Thiele Diagram [1]

$$y_{n+1} = \frac{R}{(R+1)} X_n + \frac{1}{(R+1)} X_D$$

$$\text{Where } R = L/D$$

Distillation columns can have the feed stream at any tray. The Armfield allows for the feed to be placed at the top, bottom, or middle of the column. When the feed is input from the middle it is being run as a conventional column. When the feed is placed at the bottom of the column this

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yields a rectifying column; this is where the more volatile component is removed with the contact of the vapor with the liquid. When the feed is inserted into the bottom of the column, then it is being run as a stripping column. A stripping column is defined as a column in which the liquid is being stripped of the more volatile component by the rising vapor.

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Column efficiency is a required calculation for the lab. Equation 2 below, which is obtained from the lab manual, is used to solve for this.

$$\eta = \frac{y_n - y_{n+1}}{y_n^* - y_{n+1}} \quad (2) [1]$$

y_n^* is the composition in equilibrium with X_n

If the vapor and liquid are in equilibrium, then $y_n = y_n^*$. The efficiency line will then drop closer to the operating line on the McCabe-Thiele diagram.

The relative volatility of the reaction, equation 3, is required to solve the Fenske equation, equation 4. Relative volatility is the difference between the vapor pressures of the more volatile, and less volatile components [3]. For a binary mixture the relative volatility equation is:

$$\alpha = \frac{\left(\frac{y_i}{x_i}\right)}{\left(\frac{y_j}{x_j}\right)} = \frac{K_i}{K_j} \quad (3) [3]$$

With the relative volatility obtained it is possible to solve for the Fenske equation.

$$N = \frac{\log \left[\left(\frac{X_d}{1 - X_d} \right) \left(\frac{1 - X_b}{X_b} \right) \right]}{\log(\alpha)} \quad (4) [4]$$

4. Results

The objective of this experiment was to examine the efficiency of the binary distillation column separating ethanol and water while operating in a continuous steady state manner with two different feed inlet locations. The feed, which was a 3L 30wt% ethanol solution with 70wt% water, was varied from being placed in the middle of the column for Experiment B to then being placed at the top of the column for Experiment C. Prior to beginning the core of the experiment the calculations were made to obtain the refractometer's calibration curve which correlated the relationship between refractive index and the ethanol composition of the samples taken. The polynomial best fit line was used for the calibration curve shown below in Figure 4.1.

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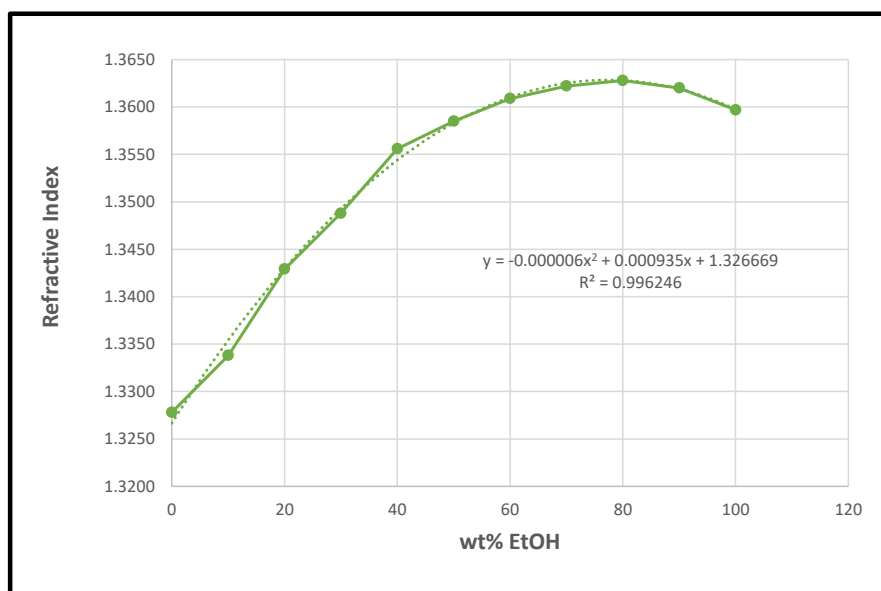


Figure 4.1: Wt% EtOH Vs. Refractive Index Curve

Experiment A: Pump Calibration

The objective of this experiment is to create a calibration curve of the pump's actual flow rate against the position of the variable speed dial on the control console. The peristaltic pump is designed to pump 1 mL/min per revolution of the drive shaft. In order to deliver an accurate flow rate into the column when running a continuous distillation, the calibration curve is needed to find exactly how much is being pumped. Figure 4.2 represents the calibration curve for the peristaltic pump that will be used in experiments B and C.

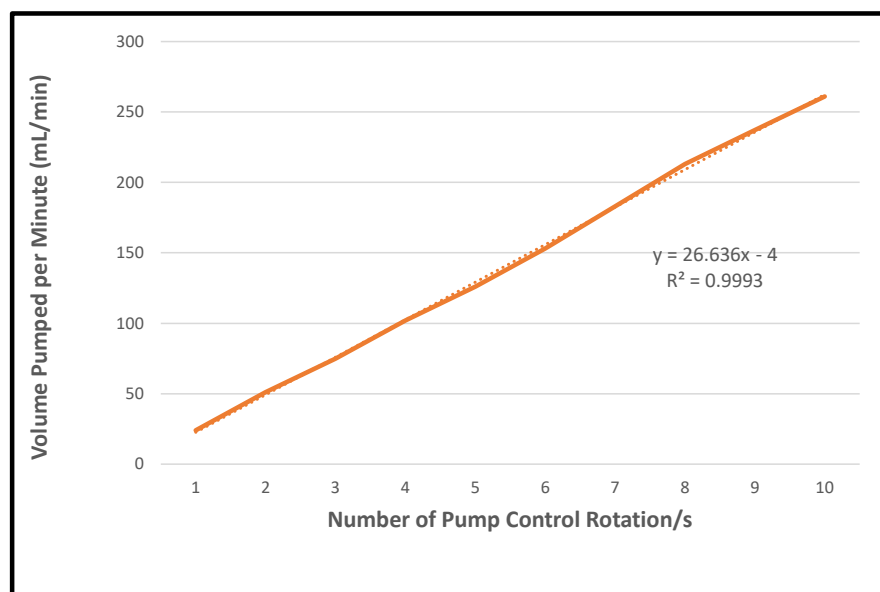


Figure 4.2: Pump Calibration (Number of Pump Control Rotation/s vs. Volume Pumped per Minute)

From the above pump calibration curve, it was found that the pump's design for the distillation system was approximately 1.46 on the variable speed dial in order to pump 35 mL/min into the column. This speed setting will be kept constant in experiments B and C when running the continuous distillation system.

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The power was initially set to 1.50kW to get the 10L reboiler solution (10wt% ethanol and 90wt%water) to the boiling point of ethanol which is approximately 78°C. Then the power was reduced as the column was approaching steady state. From Figure 4.3 below it is noted that the column progressed linearly in unsteady state until it reached the conditions of steady state after 33 minutes (sample 200). The steady condition has varying bumps since the column faced varying impediments that needed to be corrected to regain a smooth operation by adjusting the feed flow. Some notable ones are the pump's tubing having leaks and the pump not having the correct flow to pump the feed into the column at the desired rate, the error analysis portion explains more in depth the reason of each one. Here, it is seen around sample 460-500 and sample 750 the pump experienced decreases so the temperature also experienced a shift from steady state.

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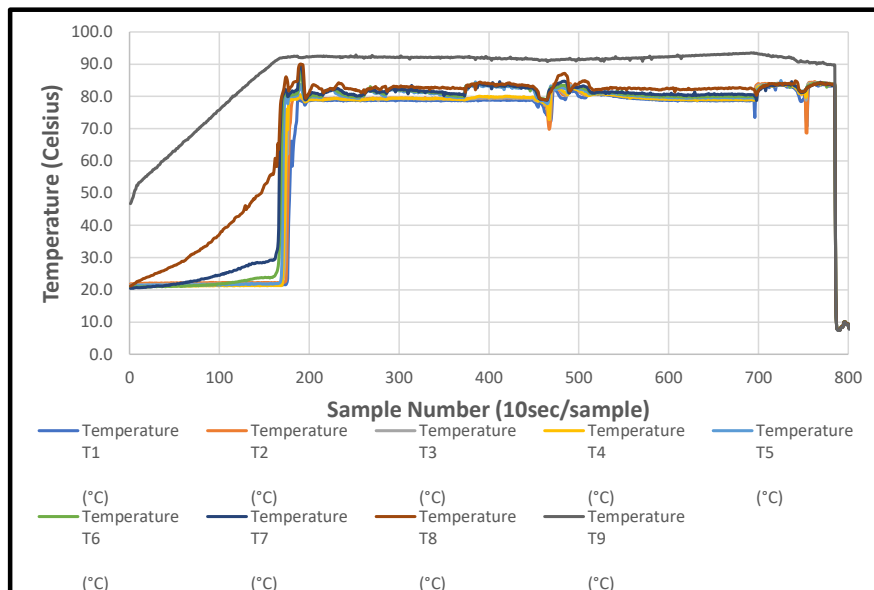


Figure 4.3: Distillation Stage Temperatures vs. Time

Experiment B

The objective of this experiment is to investigate steady state distillation under continuous operations with an incoming feed at the midpoint of the column. The power was set to 1.5kW initially to heat the boiler until constant temperatures are reached throughout the column. Once reaching steady state, the power is readjusted to 0.75kW and then the reflux ratio is set 5:1. Due to technical difficulties of the pump's lack of suction, the new setting for the speed dial was set to approximately 7 in order to gain enough flow into the column. After 500 millimeters fed, the first set of samples are taken from the top of the column and the reboiler, then another 2 sets of samples in 5-minute intervals. The refractive indexes are showed in Table 7.4, with its corresponding compositions that were calculated from the ethanol weight percent versus refractive index calibration curve. The average composition of ethanol in the distillate was 0.637, while the average composition at the bottoms was 0.136.

Once the compositions are calculated for both the top and bottom of the column, the average composition was used in determining the mole fractions of ethanol and water at each end. Mole fraction calculations were done by using the molecular weights of both components, and interpolating Table 7.8 which contains Vapor-Liquid-Equilibrium data of ethanol and water. The relative volatility of the distillate was 0.427, while the bottoms relative volatility was 8.04. Table 7.6 displays the mole fractions of ethanol and water under continuous distillation with a feed at the midpoint of the column.

Next, the theoretical number of plates was calculated using the Fenske's equation that was previously discussed above. Once knowing the theoretical and actual number of trays in the column, the column efficiency can be calculated. The column efficiency was calculated using Murphy's plate efficiency equation, which was about 36.4 %. Table 4.1 displays the estimated results of the distillation column, which included average relative volatility, theoretical plates, and the overall column efficiency.

Table 4.1: Distillation Column Efficiency Calculations for Midpoint Feed

Midpoint Feed Position	
Distillation Column Estimated Design	
Relative Volatility (Average): 1.853	Column Efficiency: 36.362
Theoretical Plates: 2.909	Actual Number of Trays: 8.000

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Experiment C

The objective of Experiment C was to analyze the steady state distillation under continuous operations with the feed being moved to the top of the column. The power was left at 0.75kW and the reflux ratio was also kept as 5:1. The pump kept having difficulties with the suction and for this experiment the leak was found and cut out so then the speed dial was set to 4 which was 102.5ml/min rate. After feeding the column about 500ml a sample was taken, from the top and bottom then after 5 minutes a second one was taking and then finally one more after another 5 minutes. The data collected is seen in Table 7.3. The refractive indexes are showed in Table 7.5, with its corresponding compositions that were calculated from the ethanol weight percent versus refractive index calibration curve. The average ethanol composition of the distillate was 0.6855 and the bottoms composition was 0.1393.

Then, Table 7.7 takes the composition averages to find the vapor and liquid mole fractions of the distillate and bottoms by utilizing interpolations of Table 7.8 Vapor-Liquid Equilibrium Data for Ethanol and Water (1atm). With obtaining the mole fractions, the relative volatility was calculated to be 2.028 as seen in Table 4.2 below. The table also displays the theoretical number of trays being equal to 2.506 and efficiency reaching 31.33%. All these calculations were done as similarly explained in the results above for Experiment B.

Table 4.2: Distillation Column Efficiency Calculations for Above Top Plate Feed

Above Top Plate Feed Position	
Distillation Column Estimated Design	
Relative Volatility (Average): 2.028	Column Efficiency: 31.330
Theoretical Plates: 2.506	Actual Number of Trays: 8.000

Figure 4.4 below displays all three trials' composition to compare the feed being placed at the top of the column versus the middle of the column. The compositions for the distillate and bottoms with varying feed location are relatively close. Although, for the distillate the composition is higher for the feed location being placed at the top. As for the bottom's composition, the midpoint feed is higher at minute 10 but then switched at minute 15 where the top plate feed is higher. Overall, the top plate feed gave an average of about 0.05 higher composition of ethanol than the midpoint feed location. Now, when comparing Tables 4.1 and 4.2 the relative volatility was 0.175 higher in the top plate feed versus the midpoint feed. The theoretical plates estimation was higher for the midpoint feed by about 0.4 and the column efficiency was 5.03% more efficient. This was because the ideal location for the feed is somewhere where the feed stream composition is like the theoretical composition of the column. So, the middle feed location was better than the top location which resulted in the higher efficiency being in the middle.

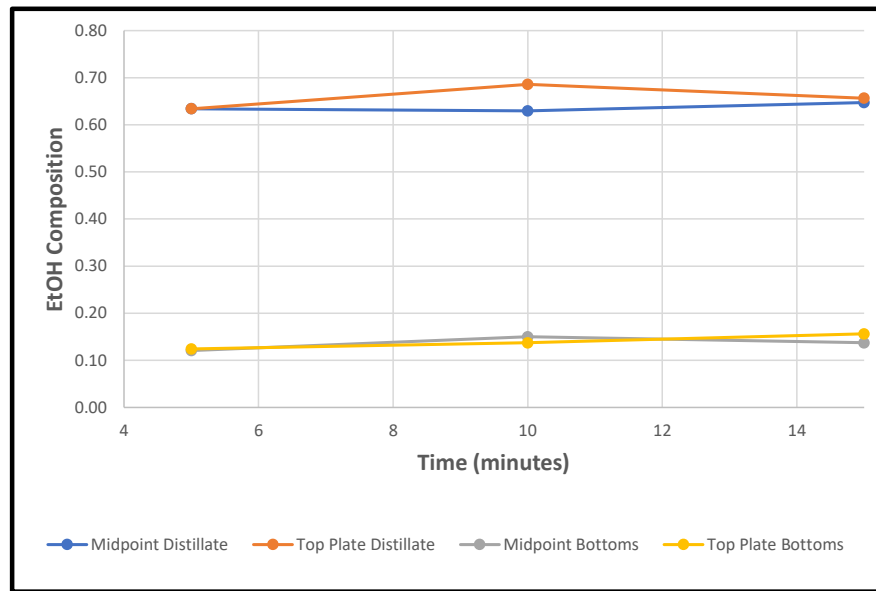


Figure 4.4: Time Vs. Ethanol Composition with Varying Feed Location

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5. Discussion

The refractive index plot of Figure 4.1, matches expected values since the refractive index of pure ethanol is 1.36 and follows the shown trend at 20C [5]. The increase in the refractive index at higher ET wt% is because of the liquid-liquid mixture causing the light from the refractometer to break up more light waves than if the solution were pure.

A McCabe Thiele analysis of distillation, in the top feed, would show that the liquid concentration of the less component would be higher at the distillate and inversely the more volatile component would have a higher concentration in the distillate. This matches our results. The relative volatility of the distillate, which is 0.6855wt% ethanol, was 0.427, while the bottoms, which is 0.136wt% ethanol, had relative volatility was 8.04.

When comparing the Feed Above the Top Plate and Feed at the Midpoint, our results do not fully match with theory. If the feed is above the Top Plate the distillation column is acting less like a distillation column and more like a stripper. Distillation contain both the stripper and rectifying column qualities. If the rectifying column aspect is removed it should be expected that a concentration of Experiment Part C should be less than ideal in comparison to Experiment Part B. This is not the case. On the other hand, the efficiency of the column operating Feed Above the Top Plate are about the same. Numerically this makes sense, since the concentrations and relative volatiles reported are very close, yet the efficiency should be significantly less for the Feed from the Top Plate.

During Experiment B and C there were many experimentally errors occurring. Many of which were since this was the first time that the lab was performed. The tubing for the feed pump had tears from the vice being too tight. This caused inconsistent flow of the feed solution and created air bubbles in the flow. Since distillation is very dependent on equilibrium of the components changing the stream into a multicomponent stream can greatly affect the results and it would be very complicated to determine how the column should have been operating.

Further experiments need to be conducted to determine the exit concentrations of the distillate and bottoms that the experiment should have yielded. It is of interest to recalculate the values to determine the efficiency and the relative volatility of the column under the two conditions. Theory would suggest the purities would be lower for the Top of the Feed.

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6. References

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- [1] Unit Operations Lab: *Continuous Distillation*. Dr. Zdunek. Fall 2019. Department of Chemical Engineering. University of Illinois at Chicago. Lab Manual.
- [2] McCabe, W. L. & Smith, J. C. (1976). *Unit Operations of Chemical Engineering (3rd ed.)*. McGraw-Hill.
- [3] Seader, J. D., and Henley, Ernest J. (1998). *Separation Process Principles*. New York: Wiley.
- [4] David S.J. Jones and Peter P. Pujado (Editors) (2006). *Handbook of Petroleum Processing*, First Edition. Springer.
- [5] Coradin, et al. "Etched Fiber Bragg Gratings Sensors for Water-Ethanol Mixtures: a Comparative Study." *Journal of Microwaves, Optoelectronics and Electromagnetic Applications*, Sociedade Brasileira De Microondas e Optoeletrônica e Sociedade Brasileira De Eletromagnetismo, www.scielo.br/scielo.php?script=sci_arttext&pid=S2179-10742010000200007.
- [6] Wankat, P., *Separation Process Engineering*. Upper Saddle River: Prentice Hall, pg. 1530. 2012.

7. Appendix I: Data Tabulation/Graphs

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Table 7.1: Experiment A-Feed Pump Calibration

Number of Pump Control Rotation/s	Volume (ml)	Time (sec)
1	24	60
2	51	60
3	75	60
4	102	60
5	126	60
6	153	60
7	183	60
8	213	60
9	237	60
10	261	60

Table 7.2: Experimental Raw Data Part B – Midpoint Feed Position

Top Product			
Sample	1	2	3
Refractive Index	1.3622	1.3621	1.3625
Sample #	382	414	447
Temperature (F)	78.8	78.8	78.8
Boiler Temp	92.1	91.6	91.6
Bottom Product			
Sample	1	2	3
Refractive Index	1.3373	1.3394	1.3391
Sample #	383	413	446
Temperature (F)	83.7	83.8	83.1

Table 7.3: Experimental Raw Data Part C – Above Top Plate Feed Position

Top Product			
Sample	1	2	3
Refractive Index	1.3622	1.3633	1.3627
Sample #	720	752	782
Temperature (F)	83.6	80.3	83.3
Boiler Temp	92.5	90.7	89.8
Bottom Product			
Sample	1	2	3
Refractive Index	1.3376	1.3393	1.3399
Sample #	720	752	782
Temperature (F)	83.6	81.7	83.7

Table 7.4: Refractive Index and Compositions at Midpoint Feed

Midpoint Feed Position						
Trial #	Refractive Index		EtOH Composition		Average Composition	
	Distillate	Bottoms	Distillate	Bottoms	Distillate	Bottoms
1	1.3622	1.3373	0.6340	0.1207	0.6370	0.1360
2	1.3621	1.3394	0.6296	0.1501		
3	1.3625	1.3391	0.6473	0.1371		

Table 7.5: Refractive Index and Compositions at Above Top Plate Feed

Above Top Plate Feed Position						
Trial #	Refractive Index		EtOH Composition		Average Composition	
	Distillate	Bottoms	Distillate	Bottoms	Distillate	Bottoms
1	1.3622	1.3376	0.6340	0.1242	0.6588	0.1392
2	1.3633	1.3393	0.6859	0.1371		
3	1.3627	1.3399	0.6565	0.1563		

Table 7.6: Ethanol Composition Summary for Midpoint Feed

Midpoint Feed Position			
Distillate Summary: Composition of Ethanol		Bottom Summary: Composition of Ethanol	
Trials Average	0.6370	Trials Average	0.1360
Xdist EtOH	0.4069	Xbot EtOH	0.0580
Xdist H ₂ O	0.5931	Xbot H ₂ O	0.9420
Ydist EtOH	0.6162	Ybot EtOH	0.3309
Ydist H ₂ O	0.3838	Ybot H ₂ O	0.6691
Relative Volatility (a)	0.4273	Relative Volatility (a)	8.0366

Table 7.7: Ethanol Composition Summary for Above Top Plate Feed

Above Top Plate Feed Position			
Distillate Summary: Composition of Ethanol		Bottom Summary: Composition of Ethanol	
Trials Average	0.6588	Trials Average	0.1392
Xdist EtOH	0.4302	Xdist EtOH	0.0595
Xdist H ₂ O	0.5698	Xdist H ₂ O	0.9405
Ydist EtOH	0.6252	Ydist EtOH	0.3371
Ydist H ₂ O	0.3748	Ydist H ₂ O	0.6629
Relative Volatility (a)	0.4526	Relative Volatility (a)	9.0915

Table 7.8: Vapor-Liquid Equilibrium Data for Ethanol and Water (1atm) [6]

X _{EtOH}	X _{water}	Y _{EtOH}	Y _{water}	Temperature °C
0.00	1.00	0.00	1.00	100
0.019	0.981	0.170	0.830	95.5
0.072	0.928	0.389	0.611	89
0.097	0.903	0.438	0.563	86.7
0.124	0.876	0.470	0.530	85.3
0.166	0.834	0.509	0.491	84.1
0.234	0.766	0.545	0.456	82.7
0.261	0.739	0.558	0.442	82.3
0.327	0.673	0.583	0.417	81.5
0.397	0.604	0.612	0.388	80.7
0.520	0.480	0.660	0.340	79.7
0.573	0.427	0.684	0.316	79.3
0.676	0.324	0.739	0.262	78.74
0.747	0.253	0.782	0.219	78.41
0.894	0.106	0.894	0.106	78.15
1.00	0.00	1.00	0.00	78.3

8. Appendix II: Error Analysis

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There were several experimental errors in lab due to the fact that it was the first time it was ever operated like a continuous distillation and the feed was in the middle and top of the column. The major source of error came from the pump and the tubing. The pump speed was not consistent with the initial value for the pump required in part A. The pump should have been run at 1.46 on the dial to obtain the optimum flowrate of 35 ml. However, the pump was run at higher speeds and then was left at 4 on the dial for a pump speed of 102.54 ml. This was due to the tube in the pump having holes in it. Another source was that the feed tube in the feedstock holder was touching the bottom of it, therefore the pump also struggled to reach the desired flow. In the last experiment, the pump tube was cut and set up again to finish the lab. Due to the pump having various problems the temperatures throughout the column had some deviations from the steady state as seen in Figure 4.3.

Since the measuring devices for the ethanol water mixture were calibrated to 1000ml the amount of ethanol added was estimated. This could have slightly deviated from the desired 10L and 3L mixtures being at exactly 10wt% and 30wt% ethanol respectively as the lab manual had required. Table 8.1 displays the standard deviation of the trials for both feed locations and that means the data is less spread out and there is less variability.

Table 8.1: Standard Deviations of the Data

	Distillate		Bottoms	
Midpoint Feed Position				
	Refractive Index	Standard Deviation	Refractive Index	Standard Deviation
Sample 1	1.3622	5E-05	1.3373	0.000927362
Sample 2	1.3621		1.3394	
Sample 3	1.3625		1.3391	
Average	1.3623		1.3386	
Above Top Plate Feed Position				
Sample 1	1.3622	0.00055	1.3376	0.000974109
Sample 2	1.3633		1.3393	
Sample 3	1.3627		1.3399	
Average	1.3627		1.3389	

9. Appendix III: Sample Calculations

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Volume of Solution (10L with 10wt% of Ethanol)

Density of Ethanol: .789 kg/L Density of Water: .997 kg/L

$$\frac{X_{EtOH}}{.787 \text{ kg/L}} + \frac{Y_{H2O}}{.997 \text{ kg/L}} = 10 \text{ L}$$

Desired Concentration of Ethanol is 10%

$$\frac{X}{X+Y} = .1$$

$$X = .111 * Y$$

$$\frac{.111Y}{.787 \text{ kg/L}} + \frac{Y_{H2O}}{.997 \text{ kg/L}} = 10 \text{ L}$$

$$Y_{H2O} = 8.74 \text{ kg}$$

$$X_{EtOH} = .9714 \text{ kg}$$

kg to Liters

$$\frac{8.74 \text{ kg H}_2\text{O}}{.997 \text{ kg/L}} = 8.76 \text{ L of H}_2\text{O}$$

$$\frac{.9714 \text{ kg EtOH}}{.787 \text{ kg/L}} = 1.24 \text{ L of EtOH}$$

Volume of Solution (3L with 30wt% of ethanol)

The same steps apply for obtaining the Volume of each respective solution in the 3L 30wt% of ethanol mix to obtain the following:

1.944L of H₂O

1.0556 L of EtOH

The following calculations were done using the Midpoint Feed Trials

Relative Volatility (α_{Dist}):

Variables:

Relative Volatility: α Vapor Mole Fraction H₂O = y_{H2O} Liquid Mole Fraction H₂O = x_{H2O}

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Vapor Mole Fraction EtOH = y_{EtOH}

Liquid Mole Fraction EtOH = x_{EtOH}

$$\alpha_{Dist} = \frac{\frac{x_{EtOH}}{y_{EtOH}}}{\frac{x_{H_2O}}{y_{H_2O}}}$$

$$\frac{\left(\frac{0.4069}{0.6162}\right)}{\left(\frac{0.5931}{0.3838}\right)} = 0.427 = \alpha_{Dist}$$

Relative Volatility (α_{Bot}):

$$\alpha_{Bot} = \frac{\frac{y_{EtOH}}{x_{EtOH}}}{\frac{y_{H_2O}}{x_{H_2O}}}$$

$$\frac{\left(\frac{0.3309}{0.0580}\right)}{\left(\frac{0.6691}{0.9420}\right)} = 8.037 = \alpha_{Bot}$$

Average Relative Volatility (α_{Avg}):

$$\alpha_{Avg} = \sqrt{\alpha_{Dist} * \alpha_{Bot}}$$

$$\alpha_{Avg} = \sqrt{0.427 * 8.037}$$

$$\alpha_{Avg} = 1.853$$

Theoretical Plates (n):

$$n + 1 = \frac{\log \left[\left(\frac{x_{EtOH}}{x_{H_2O}} \right)_{Dist} \left(\frac{x_{H_2O}}{x_{EtOH}} \right)_{Bot} \right]}{\log \left(\alpha_{\frac{EtOH}{H_2O}} \right)_{Avg}}$$

$$n + 1 = \frac{\log \left[\left(\frac{0.4069}{0.5931} \right)_{Dist} \left(\frac{0.0580}{0.9420} \right)_{Bot} \right]}{\log(1.853)}$$

$$n = 2.909 \text{ Theoretical Trays}$$

Distillation Tower Efficiency:

$$E = \frac{\text{number of theoretical plates}}{\text{number of actual plates}} * 100\%$$

$$E = \frac{2.909}{8} * 100\%$$

$$E = 36.62\%$$

10. Appendix IV: Individual Team Contributions

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8 Hours	Team
Pre-Lab Editing	1 Hour	Emir
Experimental Objective (Prelab)	0.5 Hour	Emir
Apparatus (prelab)	1 Hour	Evelyn
Materials & Supplies (prelab)	1 Hour	Mike and Evelyn
Procedure (prelab)	2 Hours	Emir and Ed
Project Safety Evaluation (prelab)	1 Hour	Ed
Final Lab Editing	2 Hour	Emir and Evelyn
Executive Summary (final lab)	2 Hours	Erik and Emir
Introduction (Final lab)	1 Hour	Erik
Theory (Final Lab)	2 Hours	Mike
Results (final lab)	4 Hours	Evelyn and Emir
Discussion (final lab)	4 Hours	Ed
References (final and/or prelab)	0.5 Hour	Team
Data Tabulation/Graphs (final)	6 Hours	Evelyn and Emir
Error Analysis (final lab)	1 Hour	Mike
Sample Calculations (final Lab)	1 Hour	Evelyn and Emir
Power Point Presentation	2 Hours	Team
Total	43 Hours	